

Top 20

PROJECT Level Interview Questions

Architecture

Data Migration

IICS · dbt · Snowflake

SENIOR DATA ENGINEER

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Q01

PROJECT OVERVIEW

Tell me about the most complex data engineering project you have worked on. What was the business problem, your role, and the outcome?

HOW TO ANSWER

1 Cover the business context clearly not just tech

2 Quantify the outcome: cost saved, time reduced, accuracy improved

3 State your specific role avoid saying 'we'

Q02

PROJECT OVERVIEW

What was the total data volume — number of tables, approximate row counts, and size in GB or TB?

HOW TO ANSWER

1 Always have exact numbers ready interviewers probe this

2 Example: 120 tables, 2B rows, ~3.5TB uncompressed

3 Mention data growth rate too static vs growing volume

Q03

PROJECT OVERVIEW

What was the data frequency one-time historical load, daily batch, near real-time, or streaming? Who defined that requirement?

HOW TO ANSWER

1 Define the SLA clearly: e.g., data must be available by 6AM IST

2 Mention if you challenged or refined the requirement

3 Distinguish between load frequency and reporting freshness

Q04

ARCHITECTURE

Walk me through the end-to-end architecture of your pipeline from source to target.

HOW TO ANSWER

1 Use Medallion layers: Bronze → Silver → Gold

2 Name every tool: ingestion, orchestration, transformation, serving

3 Mention failure handling and monitoring at each layer



Q05

TECHNOLOGY DECISIONS

**Why did you choose IICS over alternatives like ADF, Glue, or custom Python?
What were the deciding factors?**

HOW TO ANSWER

1 Show you evaluated alternatives don't say 'it was already decided'

2 Factors: native connectors, team skillset, licensing, support

3 Mention trade-offs honestly – no tool is perfect

Q06

TECHNOLOGY DECISIONS

Why Snowflake over Redshift, BigQuery, or Databricks for your target platform?

HOW TO ANSWER

1 Key differentiators: zero-copy cloning, time travel, separation of compute/storage

2 Mention multi-cluster warehouse for concurrency if relevant

3 Cost model: pay-per-second vs always-on

Q07

TECHNOLOGY DECISIONS

What orchestration tool did you use and how did you manage task dependencies?

HOW TO ANSWER

1 Name the tool: Airflow, IICS Taskflows, ADF, dbt Cloud, etc.

2 Explain DAG structure and how failures were handled

3 Mention retry logic, SLA alerting, and on-call process

Q08

DATA MODELLING

What data modelling approach did you follow Star Schema, Data Vault, or one Big tables? Who made that call and why?

HOW TO ANSWER

1 Justify the choice based on business use case

2 Star Schema for BI/reporting; Data Vault for auditability

3 Mention who was involved in the decision solo or collaborative

Q09

DATA MODELLING

How did you handle SCD Type 1 and Type 2? Walk me through the exact implementation.

HOW TO ANSWER

1 Type 1 = overwrite, Type 2 = new row with validity dates

2 In dbt: use Snapshots for Type 2 automatically

3 In Snowflake: MERGE for Type 1, Streams + MERGE for Type 2

Q10

DATA MODELLING

How did you generate surrogate keys in the target warehouse?

HOW TO ANSWER

1 Options: IDENTITY columns, MD5/SHA hash of business keys, dbt_utils.surrogate_key()

2 Hash keys are portable and deterministic preferred for migrations

3 Avoid using source system primary keys as surrogate keys

Q11

DATA QUALITY

What did source data profiling reveal before migration started? Were there any surprises?

HOW TO ANSWER

1 Always mention real findings: NULLs, duplicates, encoding issues, wrong types

2 Quantify: e.g., 12% of email columns had invalid format

3 Explain how findings changed your design decisions



Q12

DATA QUALITY

What was your reconciliation strategy row counts, checksums, aggregates, or sample audits?

HOW TO ANSWER

1 Layer your validation: count → aggregate → sample → business rule

2 Use Snowflake MINUS queries for exact row-level comparison

3 dbt tests: not_null, unique, relationships, accepted_values



Q13

DATA QUALITY

What was your accepted error threshold before go-live? Who made that decision?

HOW TO ANSWER

1 This shows business acumen — it's not always 0% error

2 Example: <0.1% row variance acceptable for historical data

3 Always involve business SMEs and document sign-off formally



Q14

PERFORMANCE

How did you optimize performance for large table loads partitioning, parallelism, bulk load, or file splitting?

HOW TO ANSWER

1 COPY INTO + S3 staging is the fastest Snowflake bulk load pattern

2 Split large files to 100-250MB for parallel COPY INTO

3 Use multi-cluster warehouse to handle concurrent loads



Q15

INCREMENTAL LOAD

How did you implement incremental loads timestamp-based, sequence-based, or CDC? What were the limitations?

HOW TO ANSWER

1 Timestamp: simple but misses deletes and back-dated updates

2 CDC via IICS PowerExchange: captures all DML from transaction logs

3 Always store last_run_timestamp in a control table – not hard-coded

Q16

INCREMENTAL LOAD

How did you handle late-arriving data and multiple updates within the same batch window?

HOW TO ANSWER

1 Late arrivals: reprocess with a lookback window (e.g., last 3 days)

2 Multiple updates in same batch: take the latest by updated_at

3 Document your late arrival SLA: data accepted up to T-2 days

Q17

ERROR HANDLING

Give me a real example of a production pipeline failure. What broke, how did you find it, and what changed afterwards?

HOW TO ANSWER

1 Use STAR format: Situation → Task → Action → Result

2 Be specific – vague answers lose credibility

3 Always end with what you changed: monitoring, design, process



Q18

ERROR HANDLING

How did you ensure pipelines were idempotent safely re-runnable without duplicates?

HOW TO ANSWER

1 Use MERGE instead of INSERT everywhere possible

2 Control tables track processed batches — prevent re-processing

3 dbt incremental models with unique_key handle this natively



Q19

CUTOVER

Walk me through your go-live cutover plan step by step. What was your rollback strategy?

HOW TO ANSWER

1

Create Snowflake zero-copy CLONE before cutover – instant rollback

2

Freeze source writes → final delta sync → switch consumers → monitor

3

Test rollback in UAT before production – never assume it will work

Q20

LEADERSHIP

Looking back what is the one technical decision you would change if you could do the project again? Why?

HOW TO ANSWER

1 Pick something real — a generic answer fails this question

2 Shows self-awareness and maturity — highly valued at senior level

3 Frame it as a learning: what you'd do differently and why





Singapore 🐼

Thank You!

Hope these 20 questions help you crack your Senior DE interview!

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 - IICS · dbt · Snowflake
 - Data Quality & Validation
 - Cutover & Rollback
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